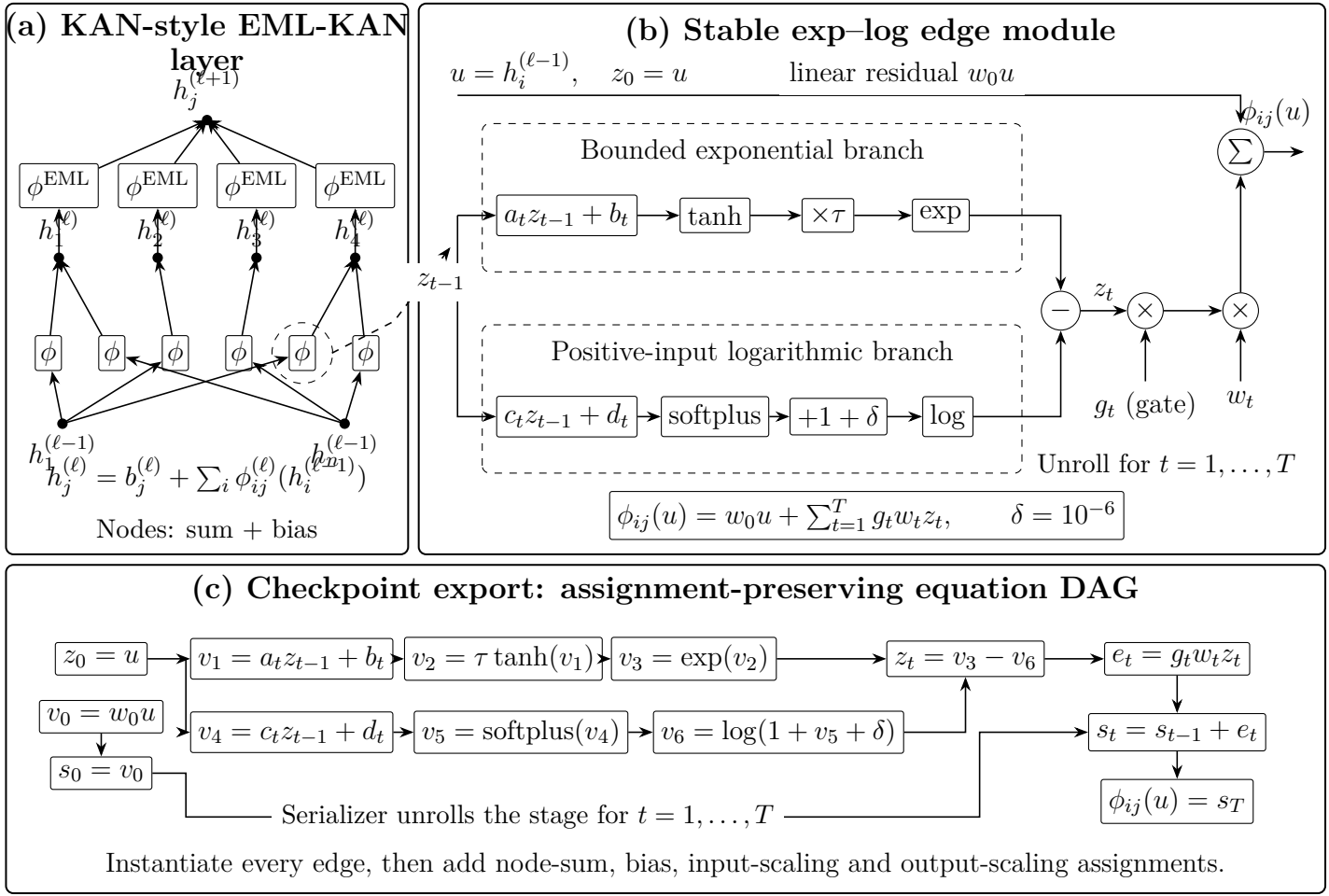

(a) KAN-style EML-KAN layer
$h_j^{(\ell+1)}$
ϕ^{EML}
$h_1^{(\ell)}$
$h_2^{(\ell)}$
$h_3^{(\ell)}$
$h_4^{(\ell)}$
ϕ
$h_1^{(\ell-1)}$
$h_m^{(\ell-1)}$
$h_j^{(\ell)} = b_j^{(\ell)} + \sum_i \phi_{ij}^{(\ell)}(h_i^{(\ell-1)})$
Nodes: sum + bias
(b) Stable exp–log edge module
$u = h_i^{(\ell-1)}, \quad z_0 = u$
linear residual $w_0 u$
$\phi_{ij}(u)$
$\sum$
Bounded exponential branch
$a_t z_{t-1} + b_t$
tanh
$\times\tau$
exp
z_{t-1}
z_t
$-$
$\times$
$\times$
Positive-input logarithmic branch
$c_t z_{t-1} + d_t$
softplus
$+1+\delta$
log
g_t (gate)
w_t
Unroll for $t = 1, \ldots, T$
$\phi_{ij}(u) = w_0 u + \sum_{t=1}^{T} g_t w_t z_t, \qquad \delta = 10^{-6}$
(c) Checkpoint export: assignment-preserving equation DAG
$z_0 = u$
$v_1 = a_t z_{t-1} + b_t$
$v_2 = \tau \tanh(v_1)$
$v_3 = \exp(v_2)$
$z_t = v_3 - v_6$
$e_t = g_t w_t z_t$
$v_0 = w_0 u$
$v_4 = c_t z_{t-1} + d_t$
$v_5 = \mathrm{softplus}(v_4)$
$v_6 = \log(1 + v_5 + \delta)$
$s_t = s_{t-1} + e_t$
$s_0 = v_0$
$\phi_{ij}(u) = s_T$
Serializer unrolls the stage for $t = 1, \ldots, T$
Instantiate every edge, then add node-sum, bias, input-scaling and output-scaling assignments.